CSA0919 PROGRAMMING IN JAVA

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1

P1.java

```java
import java.util.Scanner;
public class P1 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
*
**
***
****
*****

=== Code Execution Successful ===
```

2

P2.java

```java
import java.util.Scanner;
public class P2 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++)
                System.out.print(j+" ");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

=== Code Execution Successful ===
```

3

P3.java

```java
import java.util.Scanner;
public class P3 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter size: ");
        int n=sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=n;j++){
                if(i==1||i==n||j==1||j==n)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter size: 5
*****
*   *
*   *
*   *
*****

=== Code Execution Successful ===
```

4

P4.java

```java
import java.util.Scanner;
public class P4 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            char ch='A';
            for(int j=1;j<=i;j++){
                System.out.print(ch+" ");
                ch++;
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
A
A B
A B C
A B C D
A B C D E

=== Code Execution Successful ===
```

5

P5.java

```java
public class P5 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int n=sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int s=1;s<=n-i;s++)
                System.out.print(" ");
            for(int j=1;j<=2*i-1;j++)
                System.out.print("*");
            System.out.println();
        }

        for(int i=n-1;i>=1;i--){
            for(int s=1;s<=n-i;s++)
                System.out.print(" ");
            for(int j=1;j<=2*i-1;j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *

=== Code Execution Successful ===
```

6

P6.java

```java
import java.util.Scanner;
public class P6 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();
        int num=1;

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++)
                System.out.print(num+++" ");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

=== Code Execution Successful ===
```

7

P7.java

```java
import java.util.Scanner;
public class P7 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int s=1;s<=rows-i;s++)
                System.out.print(" ");
            for(int j=1;j<=i;j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
    *
   **
  ***
 ****
*****

=== Code Execution Successful ===
```

8

P8.java

```java
import java.util.Scanner;
public class P8 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=rows;i>=1;i--){
            for(int j=1;j<=i;j++)
                System.out.print(j+" ");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

=== Code Execution Successful ===
```

9

P9.java

```java
import java.util.Scanner;
public class P9 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++)
                System.out.print((i+j)%2+" ");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
0
1 0
0 1 0
1 0 1 0
0 1 0 1 0

=== Code Execution Successful ===
```

10

P10.java

```java
import java.util.Scanner;
public class P10 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int n=sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++)
                System.out.print("*");
            System.out.println();
        }

        for(int i=n-1;i>=1;i--){
            for(int j=1;j<=i;j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
*
**
***
****
*****
****
***
**
*

=== Code Execution Successful ===
```

11

P11.java

```java
import java.util.Scanner;
public class P11 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++){
                if(j==1||j==i||i==rows)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 3
*
**
***

=== Code Execution Successful ===
```

12

P12.java

```java
import java.util.Scanner;
public class P12 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=rows;i>=1;i--){
            for(int s=0;s<rows-i;s++)
                System.out.print(" ");
            for(int j=1;j<=2*i-1;j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
*********
 *******
  *****
   ***
    *

=== Code Execution Successful ===
```

13

P13.java

```java
import java.util.Scanner;
public class P13 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();
        char ch='A';

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++){
                System.out.print(ch+" ");
                ch++;
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
A
B C
D E F
G H I J
K L M N O

=== Code Execution Successful ===
```

14

P14.java

```java
import java.util.Scanner;
public class P14 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++)
                System.out.print(2*j+" ");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10

=== Code Execution Successful ===
```

15

P15.java

```java
import java.util.Scanner;
public class P15 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();
        int num=1;

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=i;j++){
                System.out.print(num+" ");
                num+=2;
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29

=== Code Execution Successful ===
```

16

P16.java

```java
import java.util.Scanner;
public class P16 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int s=1;s<=rows-i;s++)
                System.out.print(" ");
            for(int j=1;j<=2*i-1;j++){
                if(j==1||j==2*i-1||i==rows)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
    *
   * *
  *   *
 *     *
*********

=== Code Execution Successful ===
```

17

P17.java

```java
import java.util.Scanner;
public class P17 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter rows: ");
        int rows=sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int s=1;s<=rows-i;s++)
                System.out.print(" ");
            for(int j=1;j<=i;j++)
                System.out.print(j+" ");
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
    1
   1 2
  1 2 3
 1 2 3 4
1 2 3 4 5

=== Code Execution Successful ===
```